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Mechanism for connecting and disconnecting cluster RF connector

Abstract

A radio frequency (RF) cluster port having a port body including a number of apertures, each configured to hold a corresponding one of a number of RF ports, each RF port having an RF port conductor combination. The port body may include a number of dual-hook structures, each dual hook structure having an upper first hook and a lower second hook, wherein the upper first hook is configured to have a first pressure applied to it by a bearing pin of a cluster connector when the cluster connector engages the number of RF port conductor combinations to a corresponding number of RF connector conductor combinations, and wherein the lower second hook is configured to have a second pressure applied to it by the bearing pin of the cluster connector when the cluster connector disengages the number of RF port conductor combinations from the corresponding number of RF connector conductor combinations.

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Background/Summary

(1) This application is a continuation of U.S. patent application Ser. No. 17/737,390, filed May 5, 2022, which claims priority to U.S. Provisional Patent Application No. 63/184,306, filed May 5, 2021, all of which are hereby incorporated by reference in their entirety as if fully set forth herein.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to wireless communications, and more particularly, to cluster connectors for coupling multiple RF (Radio Frequency) cables to multiport antennas.

Related Art

(2) Modern cellular communications has experienced an explosion in demand for very high data rates per each mobile device (hereinafter user equipment or UE), as well as a massive increase in the number and types of devices. To meet the conflicting challenges of providing high data rates to an increasing number of devices, MIMO (Multiple Input Multiple Output) technologies have been developed to provide multiple simultaneous communication links between a given base station and a UE (e.g., Point-to-Point MIMO) and/or to provide spectrum reuse by enabling an antenna to establish individual narrow beams to individual UEs such that each narrow beam may use the same spectrum resources to multiple UEs simultaneously (e.g., Multi-User MIMO and Massive MIMO).

(3) Each of these approaches requires an individual antenna to have numerous radiators per supported frequency band, and numerous RF ports to provide independent RF signals to different combinations of radiators. Massive MIMO, in particular, requires a large number of RF ports. The need for an increasing number of RF ports is further complicated by the demand to reduce the size of the antenna for dense urban deployments and improved wind loading.

(4) More RF ports may be accommodated through the use of cluster connectors, in which four or more (for example) RF connections may be integrated within a single connector body. However, a complication arises in that each individual RF connection within a cluster connector may require considerable force, both for its connection as well as for its disconnection. The required force for connection/disconnection scales with the number of RF cables in a given cluster connector. The forces required for a single RF connection/disconnection may be as much as 15-20 lbs.

Accordingly, the total force required for connection/disconnection for a cluster connector or many RF connections may be considerable. Further, each RF connection must support 30+ GHz frequencies and be free of problems such as passive intermodulation distortion (PIM). This requires a precise RF engagement mechanism that may be susceptible to damage if excessive forces, such as lateral or torsional forces, are applied during insertion and removal of the cluster connector.

Additionally, in the case of a large number of RF ports (e.g., for a Massive MIMO antenna), it may be necessary to have multiple cluster connector ports disposed on the antenna in close proximity, thereby limiting access to each individual cluster connector for insertion and removal. This can be further complicated by the need to connect and disconnect these cluster connectors in the field, which may involve being at the top of a cell tower.

(5) Accordingly, there is a need for an RF cluster connector mechanism that provides for easy, consistent, and reliable connection and disconnection of its constituent RF conductors, whereby the cluster connector may be in close proximity to other cluster connectors on the antenna, and whereby the antenna may be mounted at the top of a cell tower.

BRIEF SUMMARY OF THE DISCLOSURE

(6) Accordingly, the present invention is directed to a Mechanism for Connecting and Disconnecting Cluster RF Connector that obviates one or more of the problems due to limitations and disadvantages of the related art.

(7) An aspect of the present disclosure involves an RF cluster connector. The connector comprises a connector body having a plurality of apertures configured to hold a corresponding one of a plurality of RF connector bodies, each RF connector body having a RF connector conductor combination; a plurality of lever arms rotatably coupled to the connector body at a corresponding pivot pin disposed on the connector body; and a plurality of draw arms, each rotatably coupled to a corresponding lever arm by an arm link pin at a proximal end, each of the plurality arms having a bearing pin disposed on a distal end, wherein each of the plurality of bearing pins are configured to

engage with a corresponding dual hook structure on a cluster port, each dual hook structure having an upper first hook and a lower second hook, wherein each bearing pin is configured to press against the upper first hook when engaging the plurality of RF connector conductor combinations to their corresponding RF port conductor combinations, and wherein each bearing pin is configured to press against the lower second hook when disengaging the plurality of RF connector conductor combinations from their corresponding RF port conductor combinations.

(8) Another aspect of the present disclosure involves an RF cluster port having a port body. The port body comprises a plurality of apertures configured to hold a corresponding one of a plurality of RF port conductor combinations; and a plurality of dual-hook structures, each dual hook structure having an upper first hook and a lower second hook, wherein the upper first hook is configured to have a first pressure applied to it by a corresponding bearing pin of a cluster connector when engaging the plurality of RF port conductor combinations to a corresponding plurality of RF connector conductor combinations, and wherein the lower second hook is configured to have a second pressure applied to it by the corresponding bearing pin of the cluster connector when disengaging the plurality of RF port conductor combinations from the corresponding plurality of RF connector conductor combinations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying figures, which are incorporated herein and form part of the specification, illustrate a mechanism for connecting and disconnecting a cluster RF connector. Together with the description, the figures further serve to explain the principles of the mechanism for connecting and disconnecting a cluster RF connector described herein and thereby enable a person skilled in the pertinent art to make and use the mechanism for connecting and disconnecting a cluster RF connector.
- (2) FIG. 1 is a cutaway view of an exemplary cluster connector fully engaged with a cluster port according to the disclosure.
- (3) FIG. 2 is a cutaway view of an exemplary cluster connector with its clamp mechanism in the engaged position, according to the disclosure.
- (4) FIG. 3 is a cutaway view of an exemplary cluster port according to the disclosure.
- (5) FIG. 4 is a cutaway view of an exemplary cluster connector in an initial (first) position of a sequence for coupling to an exemplary cluster port.
- (6) FIG. 5 is a cutaway view of an exemplary cluster connector in a second position of a sequence for coupling to an exemplary cluster port.
- (7) FIG. 6 is a cutaway view of an exemplary cluster connector in a third position of a sequence for coupling to an exemplary cluster port.
- (8) FIG. 7 is a cutaway view of an exemplary cluster connector in a fourth position of a sequence for coupling to an exemplary cluster port.
- (9) FIG. 8 is a cutaway view of an exemplary cluster connector in a final engaged position of a sequence for coupling to an exemplary cluster port.
- (10) FIG. 9 is a cutaway view of an exemplary cluster connector in a first position of a sequence for decoupling from an exemplary cluster port.
- (11) FIG. 10 is a cutaway view of an exemplary cluster connector in a second position of a sequence for decoupling from an exemplary cluster port.
- (12) FIG. 11 illustrates a cluster port having a number of apertures.
- (13) FIG. 12 illustrates a cluster port having a number of apertures with RF ports installed.
- ### DESCRIPTION OF EXEMPLARY EMBODIMENTS
- (14) Reference will now be made in detail to embodiments of the mechanism for connecting and

disconnecting a cluster RF connector with reference to the accompanying figures. It will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the spirit or scope of the invention. Thus, it is intended that the present invention cover the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents

(15) FIG. 1 is a cutaway view of a configuration **100** of an exemplary cluster connector **105** fully engaged with a cluster port **110** according to the disclosure. As illustrated, an RF cable **115**, which is mechanically engaged with the body of cluster connector **105** such that connector center conductor **120** is mechanically and electrically coupled with center conductor receptacle **125** of RF port **130** within cluster port **110**.

(16) Although FIGS. 1-10 illustrate a single RF connection between RF cable **115** and RF port **130** within cluster connector **105** and cluster port **110**, it will be understood that there may be multiple parallel RF connections (RF cable **115** and RF port **130**), and that one is shown for convenience of illustration. In an exemplary embodiment, cluster connector **105** and cluster port **110** may have four or five counterpart RF connections. Some of these RF connections may be identical (e.g., all having the same RF cable **115**, center conductor **120**, and RF port **130**). Alternatively, some of them may have different RF cable types (e.g., conductor diameters), and one or more may have a non-RF cable connection and may instead have a cable intended for digital communication or DC electrical power. Further, one or more may instead have a fiber optic cable, in which case the connection may be a fiber optic interface. It will be understood that such variations are possible and within the scope of the disclosure.

(17) FIG. 2 is a cutaway view of an exemplary cluster connector **105** with its clamp mechanism in the engaged position, according to the disclosure. The clamp mechanism of cluster connector **105** includes two lever arms **205**, each of which rotatably engage with the body of cluster connector **105** at pivot point **220**. Each lever arm **205** has an arm link pin **215**, which engages lever arm **205** with a corresponding draw arm **210** such that the lever arm **205** and draw arm **210** may rotate relative to each other. Each arm link pin **215** may include a torsional biasing spring (not shown) that biases the angular orientation of the corresponding draw arm **210** toward the center axis of cluster connector **105**. Each draw arm **210** has a bearing pin **225**, which is configured to engage with the cluster port **110**, which is described further below. Further, given that lever arm **205** is configured to rotate around pivot point **220**, a combination of rotations around pivot point **220** and arm link pin **215** may enable draw arm **210** to translate as well as rotate, as is described further below.

(18) The body of cluster connector **105** may mechanically engage with an RF connector body **240**, which may be held in place within the body of cluster connector **105** by a connector gasket **235**. RF connector body **240** may include an outer conductor **230**, which surrounds center conductor **120**. The center conductor **120** is configured to mechanically and electrically couple with the center conductor receptacle **125** of cluster port **110**, and the outer conductor **230** is configured to mechanically and electrically couple with the outer conductor interface **325** of cluster port **110**. As used herein, the center conductor **120** and outer conductor **230** may be referred to as an RF connector conductor combination **122**, and the center conductor receptacle **125** and outer conductor interface **325** may be referred to as an RF port conductor combination. Further as stated above, cluster connector **105** may have a plurality of RF connector bodies **240**, each coupled to a corresponding RF cable **115**.

(19) FIG. 3 is a cutaway view of an exemplary cluster port **110** according to the disclosure. Cluster port **110** has a port body that has a plurality of apertures **314** configured to hold a corresponding plurality of RF ports **130** that electrically and mechanically couple with the corresponding plurality of RF connector conductor combinations. Apertures **314** can be seen in FIG. 11 (without RF ports installed) and in FIG. 12 (with RF ports installed). Cluster port **110** has an interface gasket **330**, which may be in the form of a ring and is configured to compress as it engages with cluster

connector **105** when being coupled and decoupled. Cluster port **110** has a plurality of dual hook structure **305**, each corresponding to a draw arm **210**, and each having an upper first hook **310** and a lower second hook **315**. Upper first hook **310** includes upper lip **312** and lower second hook **315** includes lower lip **317**. Each dual hook structure **305** is configured to engage with a corresponding bearing pin **225** in a manner described further below.

(20) Lever arms **205**, draw arms **210**, and dual hook structures **305** may be formed of metal or polymer.

(21) FIG. 4 is a cutaway view of cluster connector **105** in an initial (first) position of a sequence for coupling with cluster port **110**. As illustrated, the body of cluster connector **105** is brought into contact with the cluster port **110** such that the male interface of RF connector body **240** is aligned with outer conductor interface **325**. At this stage, the bearing pins of draw arms **210** may be in contact with the body of cluster port **110** above respective dual hook structures **305**, and the cluster connector **105** may have a remaining translation distance d to be fully engaged with cluster port **110**.

(22) FIG. 5 is a cutaway view of cluster connector **105** in a second position of a sequence for coupling to cluster port **110**. The illustrations of FIGS. 4-8 may be snapshots of a single motion made by a technician (not shown) who—at the stage depicted in FIG. 5—is rotating lever arms **205** outward from the center axis of cluster connector **105**. In doing so, the lever arms **205** rotate around their respective pivot pins **220** and thus push draw arms **210** downward as they rotate relative to the lever arms **205** around respective arm link pins **215**. As the draw arms **210** rotate and translate downward, the bearing pins **225** translate along the upper surface of the dual hook structures **305**. Due to the torsional bias provided at arm link pin **215**, bearing pins **225** are drawn toward the center axis of cluster connector **105**, and thus also toward the center axis of cluster port **110**. Further, as illustrated, at the stage depicted in FIG. 5, the remaining translation distance d is reduced as the cluster connector **105** and cluster port **110** come together.

(23) FIG. 6 is a cutaway view of cluster connector **105** in a third position of a sequence for coupling to cluster port **110**. At this stage, the technician is continuing to rotate downward lever arms **205**, which are illustrated at approximately 90 degrees from a center axis of the cluster connector **105** and cluster port **110**. With the continued rotation of lever arms **205**, draw arms **210** have translated downward to where their respective bearing pins **225** have entered the apertures of dual hook structures **305**, being drawn in by the torsional bias provided at arm link pin **215**.

(24) FIG. 7 is a cutaway view of cluster connector **105** in a fourth position of a sequence for coupling with exemplary cluster port **110**. As with the FIGS. 5 and 6, this illustration is a snapshot of a continuous motion made by an installing technician. Here, the technician is rotating lever arms **205** upward and toward the center axis of cluster connector **105**. As each lever arm **205** rotates around its respective arm link pin **215**, corresponding draw arm **210** is drawn upward such that its bearing pin **225** translated upward to where it engages with and applies pressure to upper first hook **310**. With this established, any further upward rotation of lever arms **205** causes the body of cluster connector **105** to translate downward, thereby causing outer conductor **230** to engage outer conductor interface **325**, and causing center conductor **120** to engage center conductor receptacle **125**.

(25) FIG. 8 is a cutaway view of exemplary cluster connector **105** in a final engaged position of a sequence for coupling to exemplary cluster port **110**. As illustrated, the technician has rotated lever arms **205** inward toward the center axis of cluster connector **105** until they have reached their respective resting position. The body of cluster connector **105** may have a pair of “hammer head” style tabs **250** into which the end of the lever arm **205** snaps into place. Each lever arm **205** may also have a cam structure integrated into pivot pin **220** that requires a force to be applied manually to get the lever arm **205** to be initially rotated from its neutral position illustrated in FIGS. 1, 2, 4, and 8. At this state, the RF connector body **240** is fully engaged with RF port **130**. In other words, the RF connector conductor combination has fully engaged with its corresponding RF port

conductor combination.

(26) FIG. 9 is a cutaway view of cluster connector **105** in a first position of a sequence for decoupling from cluster port **110**. As illustrated, the technician rotates lever arms **205** downward and away from the center axis of cluster connector **105**. In response, draw arms **210** rotate in turn around arm link pin **215** and translate toward cluster port **110**. This downward translation causes bearing pins of draw arms **210** to press against lower second hooks **315** of dual hook structures **305** within the body of cluster port **110**. This downward force against lower second hooks **315** causes cluster connector **105** to translate upward from cluster port **110**, thereby causing center conductor **120** to begin to decouple from center conductor receptacle **125** and outer conductor **230** to begin to decouple from outer conductor interface **325**.

(27) FIG. 10 is a cutaway view of cluster connector **105** in a second position of a sequence for decoupling from cluster port **110**. As illustrated, the technician continues to rotate the lever arms **205** downward, causing the bearing pins **225** to maintain pressure on lower second hooks **315**, thereby causing center conductor **120** to fully decouple from center conductor receptacle **125** and outer conductor **230** to fully decouple from outer conductor interface **325**.

(28) Although not illustrated, in the final motion, after center conductor **120** has decoupled from center conductor receptacle **125** and outer conductor **230** has decoupled from outer conductor interface **325**, the technician may rotate lever arms **205** upward to return them to the positions illustrated in FIG. 2.

(29) Although the term “position” is used with reference to the drawings, it will be understood that these images are snapshots of a fluid motion, and that the lever arms **105** (for example) need not be held or maintained in the positions illustrated in FIGS. 5, 6, 7, 9, and 10.

(30) In an exemplary embodiment, cluster connector **105** and cluster port **110** may support four 2.2-5 connectors. Each of these connectors may require 15-20 lbs. of force to engage and disengage. Accordingly, the total force required to engage and disengage cluster connector **105** and cluster port **110** may be 60-80 lbs.

(31) While various embodiments of the present invention have been described above, it should be understood that they have been presented by way of example only, and not limitation. It will be apparent to persons skilled in the relevant art that various changes in form and detail can be made therein without departing from the spirit and scope of the present invention. Thus, the breadth and scope of the present invention should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. An RF cluster port having a port body, the port body comprising: a plurality of apertures, each configured to hold a corresponding one of a plurality of RF (Radio Frequency) ports, each RF port having an RF port conductor combination comprising (i) a center conductor receptacle and (ii) an outer conductor interface; and a plurality of dual-hook structures, each dual hook structure having an upper first hook and a lower second hook, wherein the upper first hook is configured to have a first pressure applied to it by a bearing pin of a cluster connector when the cluster connector engages the plurality of RF port conductor combinations to a corresponding plurality of RF connector conductor combinations, and wherein the lower second hook is configured to have a second pressure applied to it by the bearing pin of the cluster connector when the cluster connector disengages the plurality of RF port conductor combinations from the corresponding plurality of RF connector conductor combinations.

2. The RF cluster port of claim 1, wherein the upper first hook comprises an upper lip, and the lower second hook comprises a lower lip, wherein the lower lip extends further from the RF port than the upper lip.

3. The RF cluster port of claim 1, wherein the plurality of RF port conductor combinations are configured to engage a single RF cable type.
 4. The RF cluster port of claim 1, wherein the plurality of RF port conductor combinations are configured to support different cable types.
 5. An RF cluster port having a port body, the port body comprising: a plurality of apertures, each configured to hold a corresponding one of a plurality of RF (Radio Frequency) ports, each RF port having an RF port conductor combination; a plurality of dual-hook structures, each dual hook structure having an upper first hook and a lower second hook, wherein the upper first hook is configured to have a first pressure applied to it by a bearing pin of a cluster connector when the cluster connector engages the plurality of RF port conductor combinations to a corresponding plurality of RF connector conductor combinations, and wherein the lower second hook is configured to have a second pressure applied to it by the bearing pin of the cluster connector when the cluster connector disengages the plurality of RF port conductor combinations from the corresponding plurality of RF connector conductor combinations; and a gasket disposed on the port body, wherein the gasket is configured to compress when the cluster connector engages the plurality of RF port conductor combinations to the corresponding plurality of RF connector conductor combinations.
 6. The RF cluster port of claim 5, wherein the upper first hook comprises an upper lip, and the lower second hook comprises a lower lip, wherein the lower lip extends further from the RF port than the upper lip.
 7. The RF cluster port of claim 5, wherein the plurality of RF port conductor combinations are configured to engage a single RF cable type.
 8. The RF cluster port of claim 5, wherein the plurality of RF port conductor combinations are configured to support different cable types.
 9. An RF cluster port having a port body, the port body comprising: a plurality of apertures, each configured to hold a corresponding one of a plurality of RF (Radio Frequency) ports, each RF port having an RF port conductor combination; a plurality of dual-hook structures, each dual hook structure having an upper first hook and a lower second hook, wherein the upper first hook is configured to have a first pressure applied to it by a bearing pin of a cluster connector when the cluster connector engages the plurality of RF port conductor combinations to a corresponding plurality of RF connector conductor combinations, and wherein the lower second hook is configured to have a second pressure applied to it by the bearing pin of the cluster connector when the cluster connector disengages the plurality of RF port conductor combinations from the corresponding plurality of RF connector conductor combinations; and wherein each of the dual-hook structures is configured to have the bearing pin translate along its outer surface when the cluster connector engages the RF cluster port and before the bearing pin applies the first pressure to its upper first hook.
 10. The RF cluster port of claim 9, wherein the upper first hook comprises an upper lip, and the lower second hook comprises a lower lip, wherein the lower lip extends further from the RF port than the upper lip.
 11. The RF cluster port of claim 9, wherein the plurality of RF port conductor combinations are configured to engage a single RF cable type.
 12. The RF cluster port of claim 9, wherein the plurality of RF port conductor combinations are configured to support different cable types.
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